



Pharmacology I

เภสัชวิทยา 1

Principle of drug toxicity

ผศ.ดร.ภก. อาณัฐชัย ม้ายอุเทศ

ภาควิชาเภสัชวิทยา คณะเภสัชศาสตร์ มหาวิทยาลัยมหิดล



วัตถุประสงค์รายวิชา

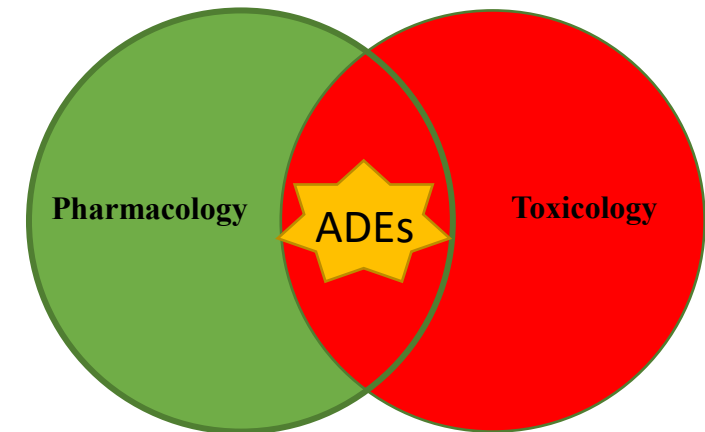
เมื่อจบการเรียนรู้ นักศึกษาสามารถ

1. สามารถบอกความแตกต่างและเลือกใช้พารามิเตอร์เพื่อบ่งชี้ความเป็นพิษของยาได้
2. สามารถบอกความแตกต่างและอธิบายกลไกการเกิดอาการไม่พึงประสงค์ อาการข้างเคียง พิษ จากการใช้ยาแบบต่างๆได้
3. สามารถอธิบายกลไกการเกิดพิษแพ้ยาชนิดต่างๆได้
4. เข้าใจกระบวนการพัฒนายาและติดตามความเป็นพิษของยาใหม่ได้



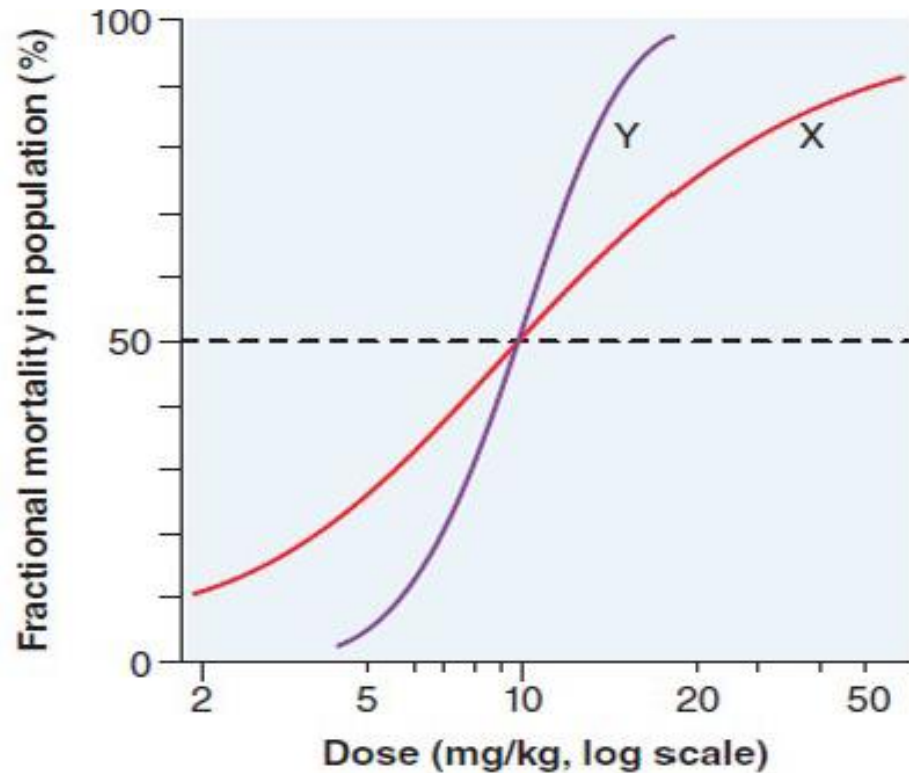
Pharmacology VS. Toxicology

- **Pharmacology** intersects with **toxicology** when the physiological response to a drug is an **adverse effect**.
 - Adverse Drug Events (ADEs)
- A poison is **any substance**, including any **drug**, that has the capacity to harm a living organism.





Conventional Dose-response curves



LD₅₀ : Median Lethal Dose

The **LD₅₀** of a compound is determined **experimentally** by administration of the chemical to mice or rats (orally or intraperitoneally).

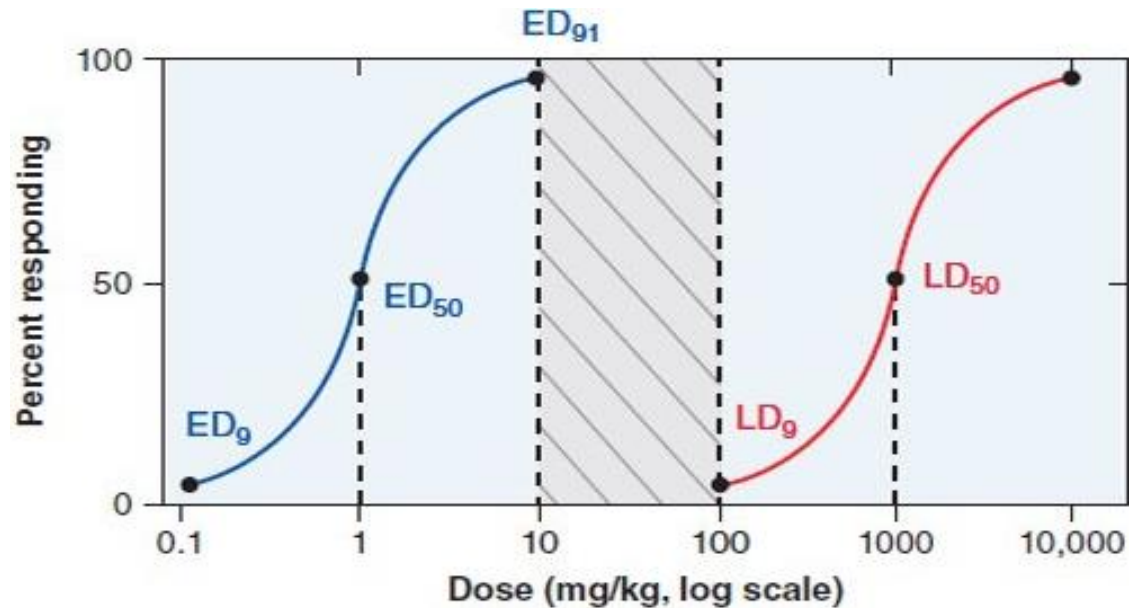
The **midpoint of the curve** representing percentage of population responding (response here is **death**) **versus dose** (log scale) represents the LD₅₀.

-The dose of drug that is lethal in 50% of the population.

The LD₅₀ values for both compounds are the same (~10 mg/kg); however, the slopes of the dose-response curves are quite different.



Margin of safety VS Therapeutic Index



The crosshatched area between the ED₉₁ (10 mg/kg) and the LD₉ (100 mg/kg) gives an estimate of the margin of safety

$$\text{Margin of safety} = \frac{LD_1}{ED_{99}}$$

Therapeutic Index (TI)

a ratio that compares the drug concentration at which causes a therapeutic effect to the amount that causes death (in animal studies) or toxicity (in human studies)

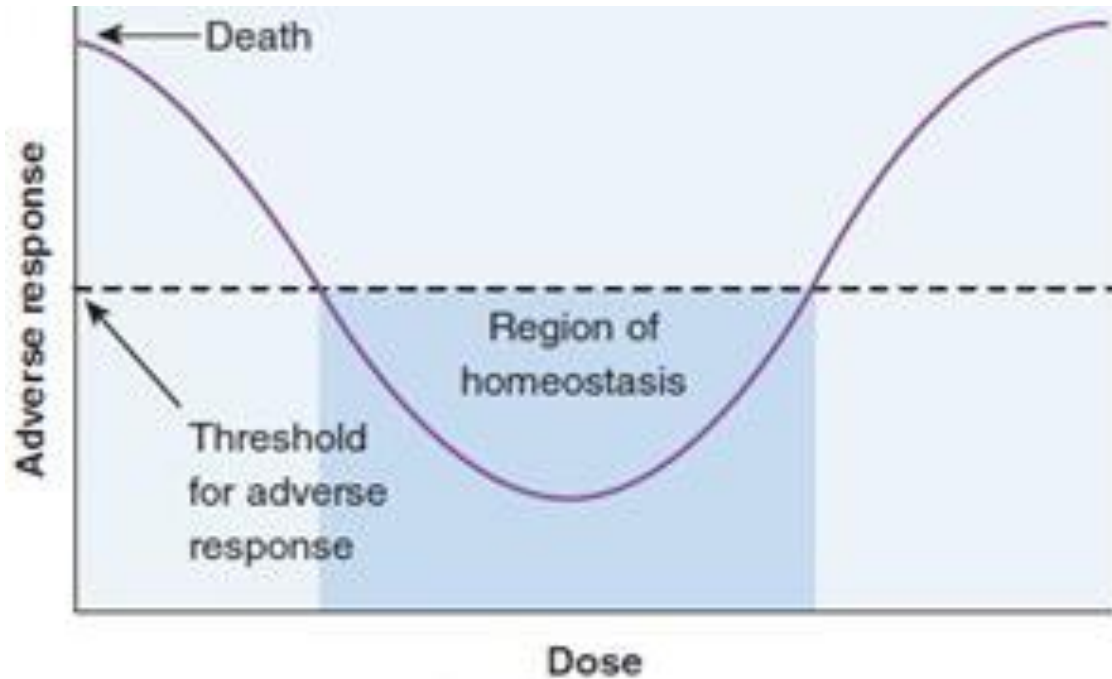
Drugs with a low TI must be administered with caution (e.g., the cardiac glycoside “digoxin” and cancer therapeutic agents).

ED₅₀ = median effective dose
LD₅₀ = median lethal dose

$$TI = \frac{LD_{50}}{ED_{50}}$$



Nonmonotonic Dose-response curves: **U-shaped**



U-shaped dose-response curve for essential metals and vitamins

At low dose, adverse effects are observed because there is a deficiency of these nutrients to maintain homeostasis.

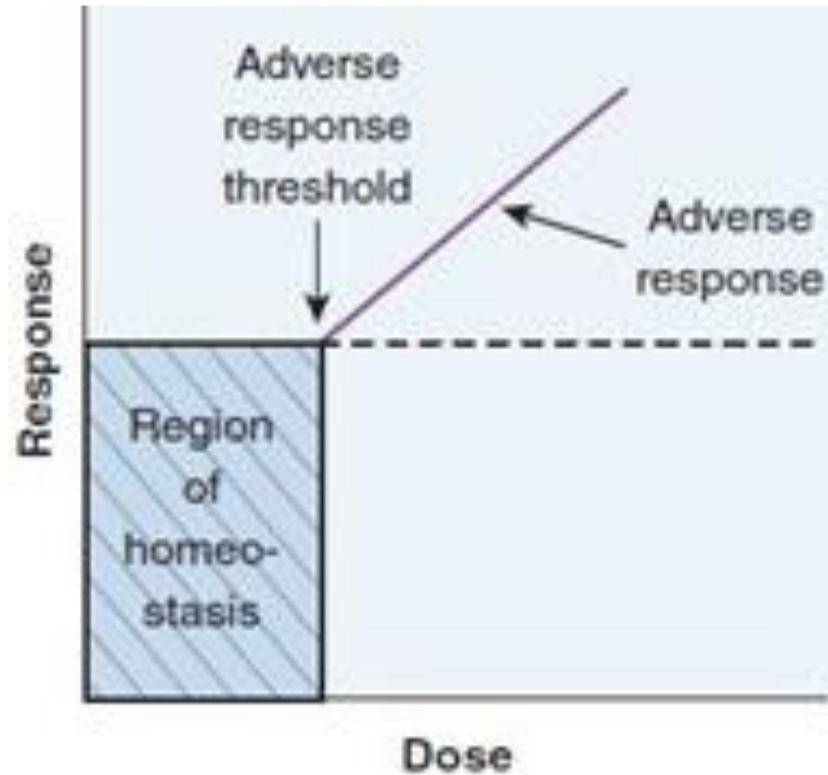
At dose that homeostasis is achieved, and the bottom of the U-shaped dose-response curve is reached.

At dose that surpass to maintain homeostasis, overdose toxicity can ensue.

Thus, adverse effects are seen at both low and high doses.



Nonmonotonic Does-response curves: **Hockey stick-shaped**



Hockey stick-shaped dose response curve for toxicants that are also metabolic by-products.

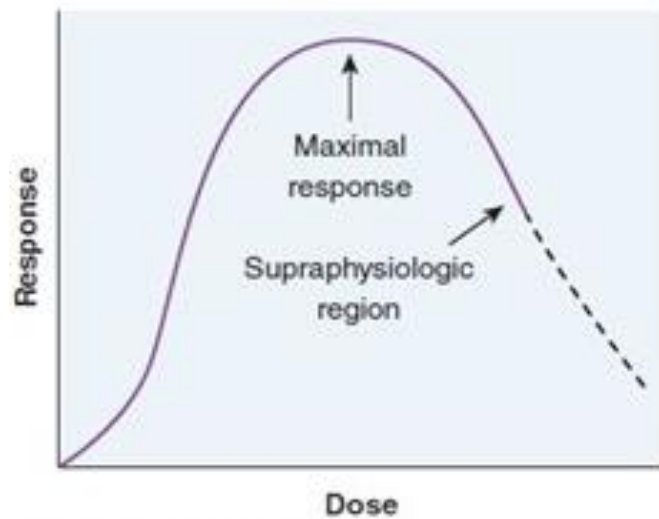
Some toxicants, such as **formaldehyde**, are also metabolic by-products for which cells have detoxifying mechanisms.

At very low doses of formaldehyde do not saturate the detoxifying mechanisms, alcohol dehydrogenase.

When these endogenous **protective mechanisms are overwhelmed**, one will observe a toxic response.



Nonmonotonic Dose-response curves: **Inverted U-shaped**



Inverted U-shaped dose-response curve for ligands that downregulate their receptors.

when **receptor downregulation/desensitization** occurs following **exposure to a ligand** or when **negative effect** occurs at the therapeutic concentration.

Estrogen at high levels can have maximal effects.

At **supraphysiologic levels**, the **effects** of estrogen **are reduced**.

Endocrine-disrupting chemicals have inverted U-shaped dose-response curves.

Multiphasic and **U-shaped curves** are **common** in an administered **compound with multiple effects**, first one effect and then another, possibly opposing, effect as the concentration increases.



Pharmacokinetic VS Toxicokinetic

The pharmacokinetics of a drug under circumstances that **produce toxicity** or **excessive exposure** are referred to as **toxicokinetics**.

-How long will an asymptomatic patient need to be monitored (**drug absorption and dynamics**)?

-How long will it take an intoxicated patient to get better (**drug elimination and dynamics**)?

Drug Absorption

Aspirin poisoning:

In **therapeutic dosing**, aspirin reaches peak plasma concentrations in about 1 h.

Aspirin overdose may cause **spasm of the pyloric valve**, **delaying** entry of the drug **into the small intestine**.

Peak plasma salicylate concentrations from aspirin overdose **may not be reached for 4–35 h after ingestion**.

Aspirin, **enteric-coated forms**, may coalesce into **bezoars**, reducing the effective surface area for absorption.

Drug Elimination

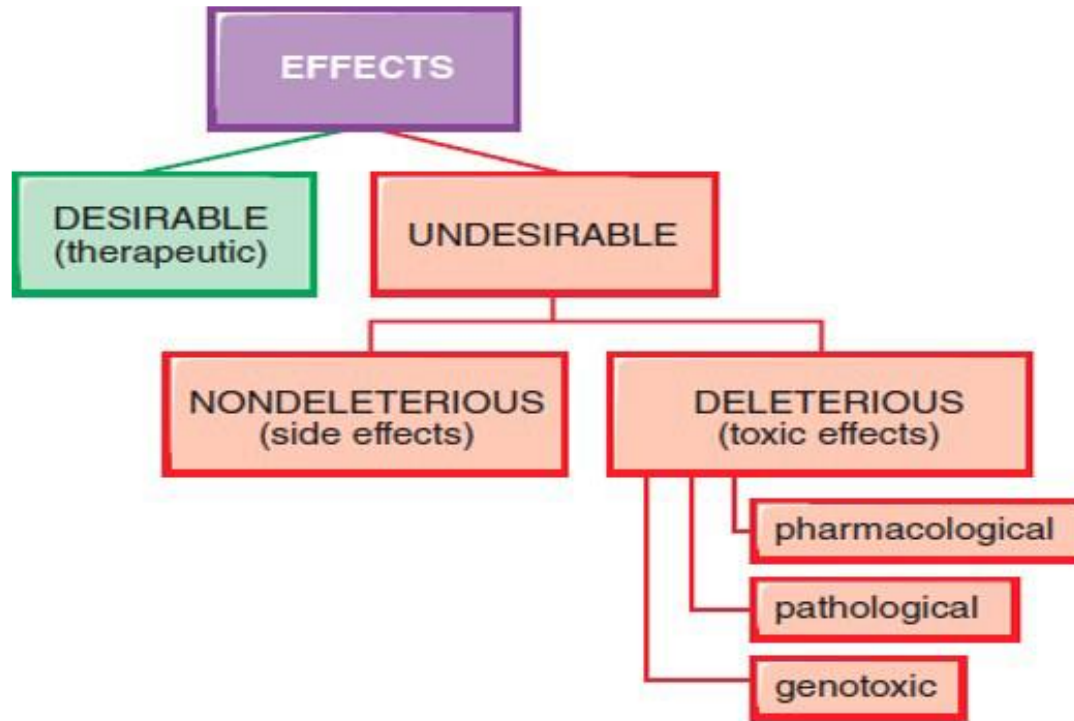
Lists some drugs to have initial symptoms develop after 4- to 6-hour of oral overdose.

- Acetaminophen
- Aspirin
- Monoamine oxidase inhibitor
- Sulfonylureas
- Thyroid hormones
- **Valproic acid**
 - after overdose, first-order metabolic processes for valproate appear to become saturated, and the apparent elimination $t_{1/2}$ may exceed 30–45 h (normal is 14h).
- Warfarin-like anticoagulants
- Sustained-released formulation drugs
- Drug in plastic package

Drugs with anticholinergic activity, diminished GI motility, may also exhibit delayed onset of action



Type of Therapeutic Drug Toxicity



•Side-effects (อาการข้างเคียงจากการใช้ยา)

- any effect caused by a drug other than the intended therapeutic effect
 - beneficial, neutral or harmful
 - Antihistamine for sedation
 - Opioids for diarrhea
- The term 'side-effect' is often used interchangeably with 'ADR'
 - Side effect usually implies an effect that is less harmful
 - Predictable
 - Lack of receptor specificity
 - Tissue distribution of the receptor
- may not even require discontinuation of therapy

•An adverse drug reaction (ADR) (อาการไม่พึงประสงค์จากการใช้ยา)

- unwanted or **harmful** reaction experienced following
 - administration of a drug
 - administrating of combination drugs (drug-drug interaction)
- Occur in normal conditions of use and is suspected to be related to the drug (not genetic).
 - the dose or plasma concentration has risen above the therapeutic range (typeA)
- usually require the drug to be discontinued or the dose reduced.



Types of ADRs

Type A

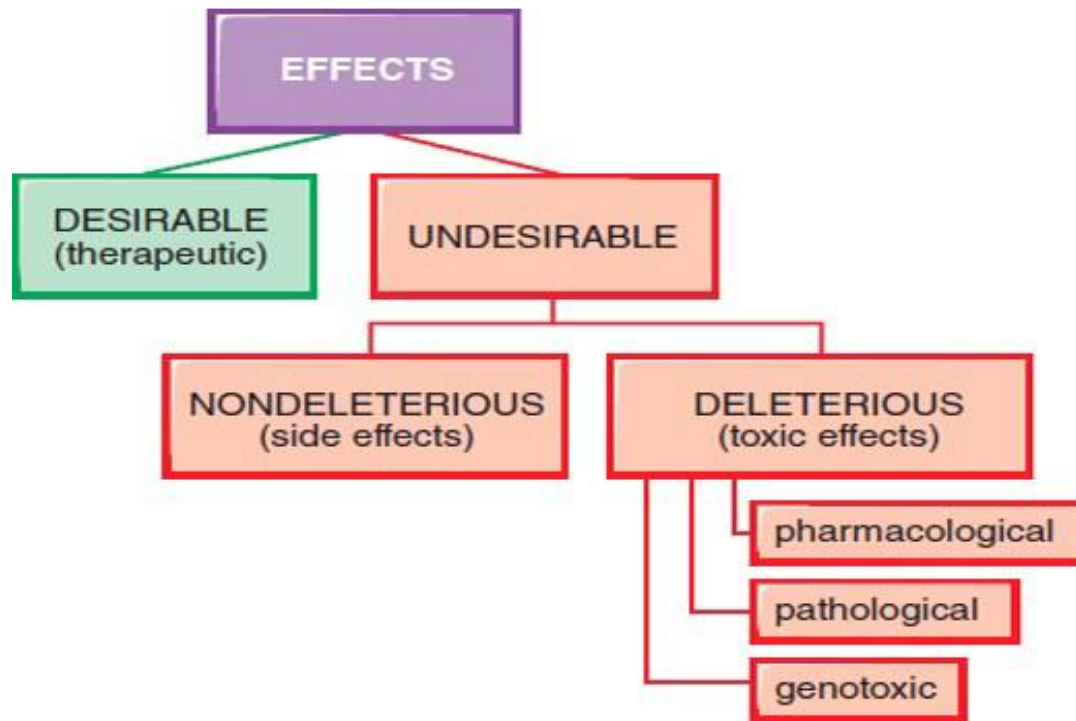
- **‘Predictable’ ADRs:**
 - Predicted from pharmacological reaction and are usually dose-related
- majority of ADRs in clinical practice
- Can be avoided
 - careful selection of drug, dose etc.

Type B:

- **‘Unpredictable’ ADRs:**
 - Not associated with pharmacology of drug
 - not dose-dependent
 - can be very serious
- May not show in clinical trials
 - low incidence
- Often involve hypersensitivity reactions



Type of Therapeutic Drug Toxicity



Pharmacological Toxicity

Toxic dose:

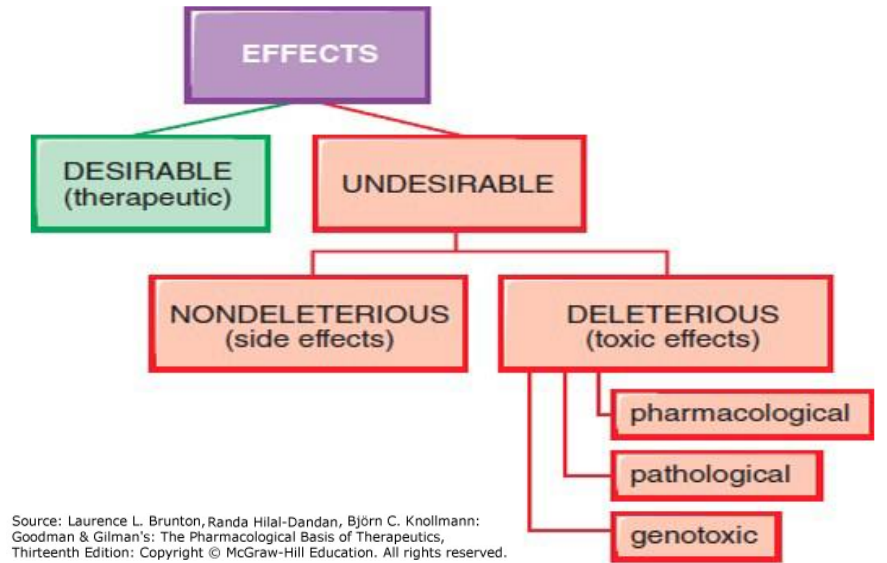
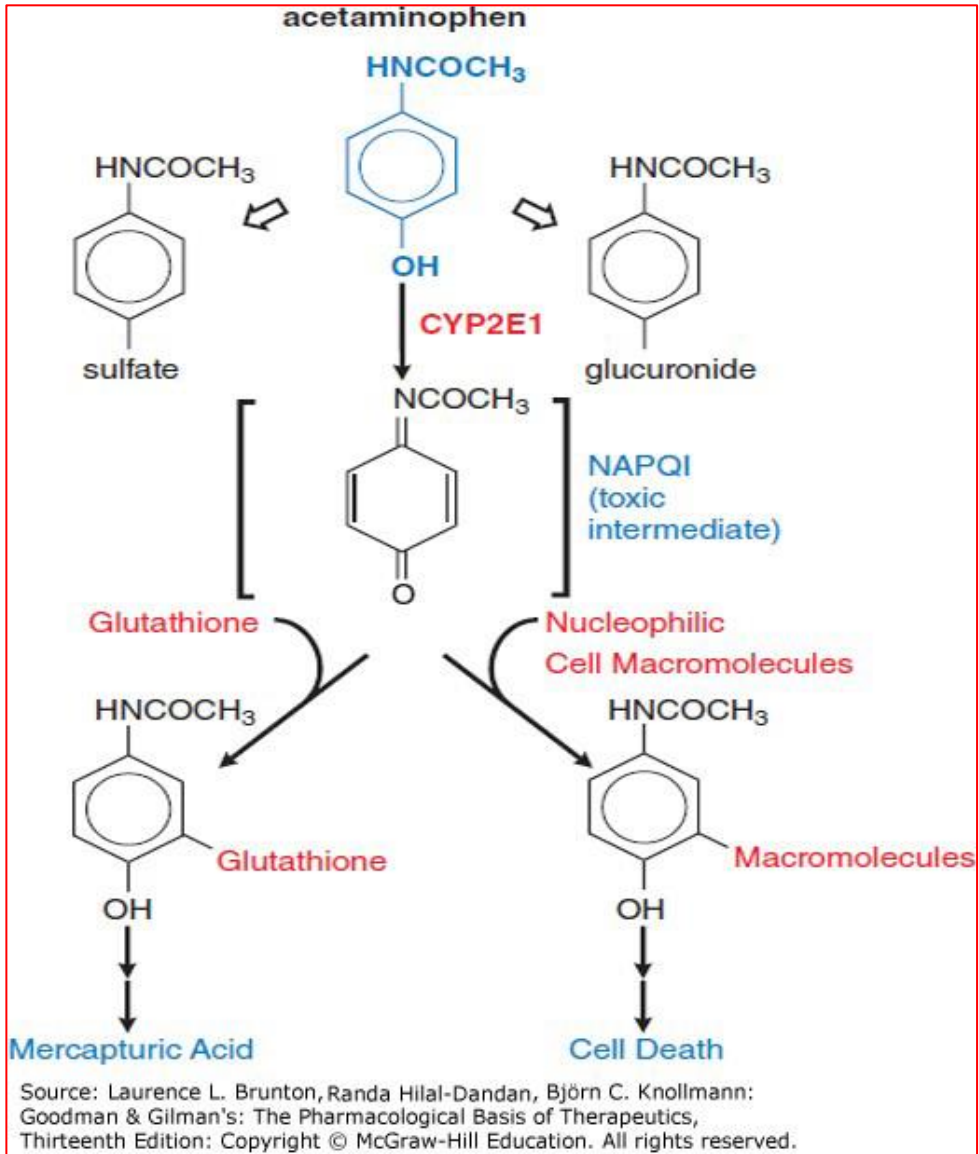
- Barbiturates produce clinical effects go from anxiolysis to sedation to somnolence to coma in **dose-dependent manner**.
- Nifedipine produced-hypotension is related to the **drug concentration**.
- Antipsychotic can cause Tardive dyskinesia (extrapyramidal disorder), depend on **duration** of exposure.

Therapeutic dose:

Tetracyclines, sulfonamides, chlorpromazine, and nalidixic acid can produce phototoxicity in the therapeutic dose.



Type of Therapeutic Drug Toxicity



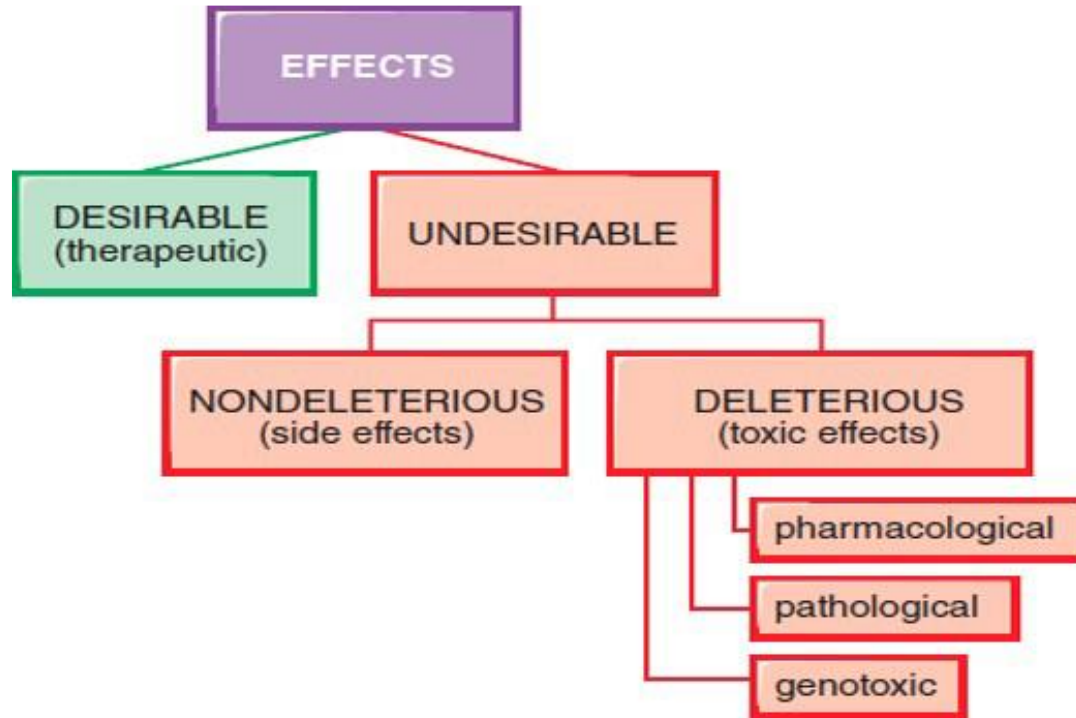
Pathological Toxicity

Acetaminophen is metabolized to nontoxic glucuronide and sulfate conjugates and to a highly reactive metabolite NAPQI via CYP isoforms. **At a therapeutic dose of acetaminophen**, NAPQI binds to nucleophilic glutathione,

At acetaminophen overdose, glutathione depletion may lead to the pathological finding of hepatic necrosis due to shunting of NAPQI toward interactions with nucleophilic cellular macromolecules.



Type of Therapeutic Drug Toxicity



Genotoxic Effects

Ionizing radiation and many environmental chemicals are known to **injure DNA** and may lead to **mutagenic** or **carcinogenic** toxicities.

Many of the cancer chemotherapeutic agents may be genotoxic.



Allergic reaction

An **allergy** is an adverse reaction, mediated by the **immune system**.

- Results from previous sensitization to a **particular chemical** or to one that is **structurally similar**.
- Allergic responses have been **divided into 4 categories** based on the mechanism of immunological.
 - Anaphylactic reaction
 - is mediated by **IgE antibodies**
 - Cytolytic reaction
 - is mediated by **both IgG and IgM antibodies** and usually are attributed to their capacity to activate the complement system
 - Arthus reaction
 - is mediated by **IgG antibody**; the mechanism involves the generation of **antigen-antibody complexes that subsequently fix complement**.
 - Delayed hypersensitivity reaction
 - is mediated by sensitized **T lymphocytes** and macrophages



Type I: Anaphylactic reaction

Anaphylaxis is mediated by IgE antibodies.

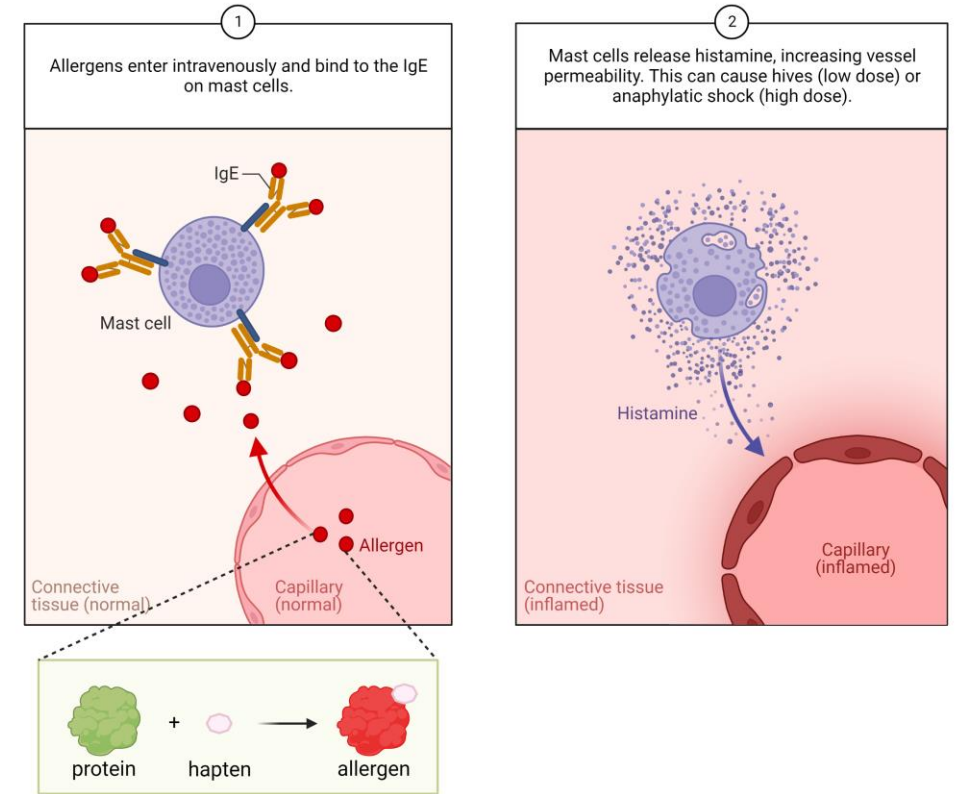
- The Fc portion of **IgE** can bind to receptors on **mast cells** and **basophils**.
- If the Fab portion of the antibody molecule **then binds an antigen**, various mediators (e.g., histamine, leukotrienes, and prostaglandins) are released and cause **vasodilation**, **edema**, and an **inflammatory response**.

The main targets of this type of reaction are

- the GI tract (food allergies)
- the skin (urticaria and atopic dermatitis)
- the respiratory system (rhinitis and asthma)
- the vasculature (anaphylactic shock)

These responses tend to **occur quickly** after challenge with an antigen to which has been sensitized

- are termed **immediate hypersensitivity reactions**





Type II: Cytolytic reaction

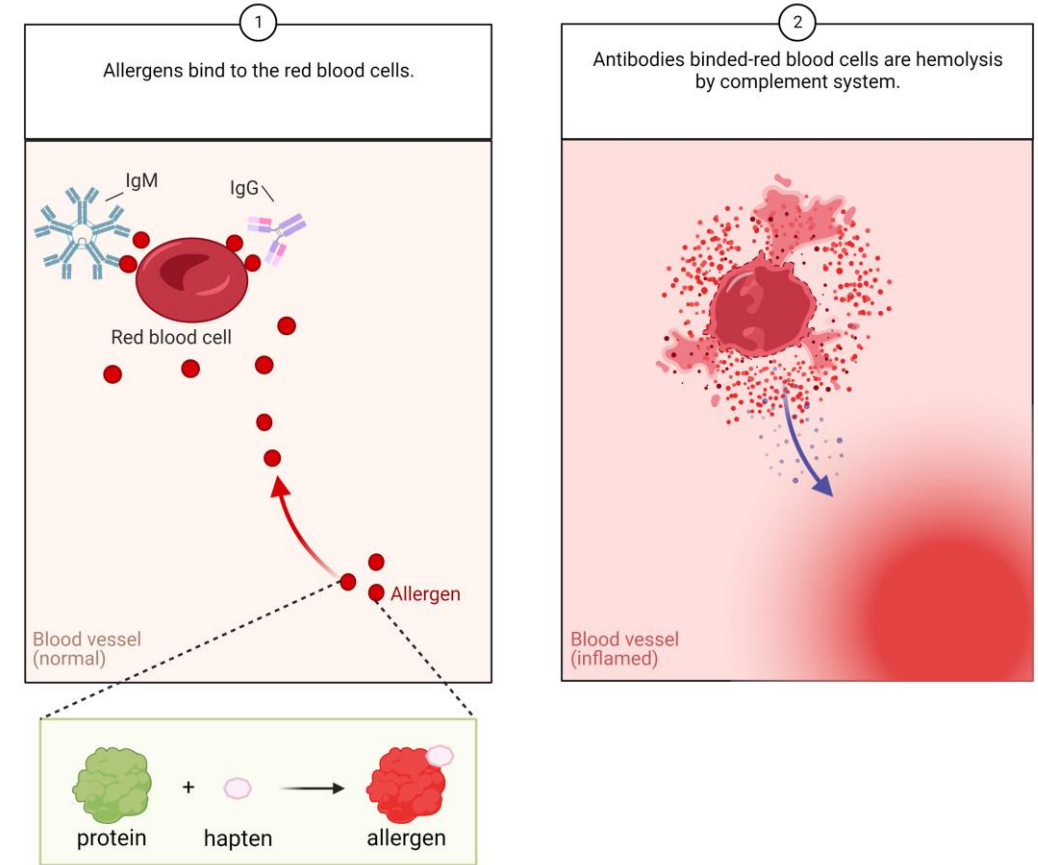
Cytolytic reaction is mediated by both IgG and IgM antibodies

- capacity to activate the **complement system**.

The major target tissues for cytolytic reactions are

- the cells in the **circulatory system**.
 - penicillin-induced hemolytic anemia
 - quinidine-induced thrombocytopenia
 - sulfonamide-induced granulocytopenia.

These autoimmune reactions to drugs usually subside within **several months** after removal of the offending agent.

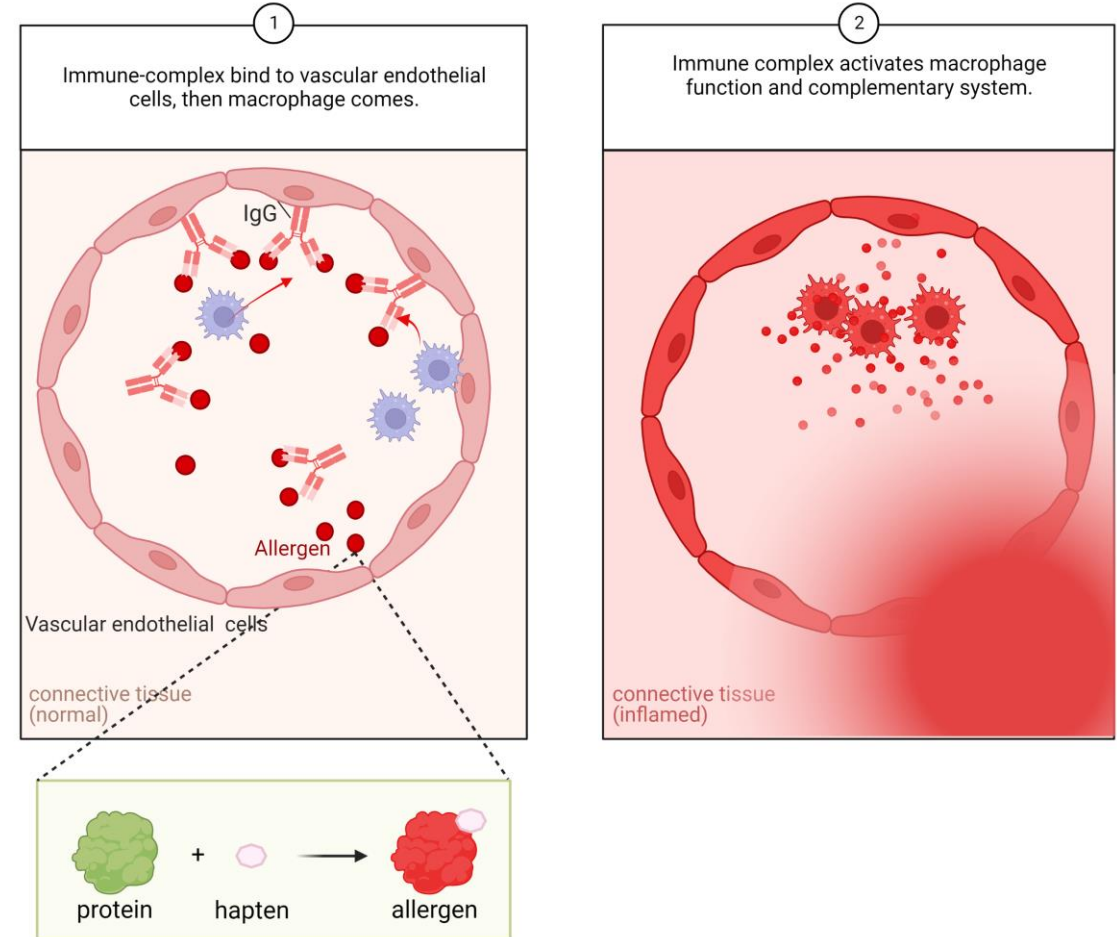




Type III: Arthus reaction

Arthus reaction is mediated predominantly by IgG

- the mechanism involves the generation of antigen-antibody complexes that subsequently **fix complement**.
- The complexes are deposited in the **vascular endothelium**, where a destructive inflammatory response called **serum sickness** occurs.
- The **clinical symptoms of serum sickness** include
 - urticarial skin eruptions
 - arthralgia or arthritis
 - lymphadenopathy
 - fever
- Several drugs, including commonly used antibiotics, can induce serum sickness-like reactions, **cefactor**, **mitomycin C**.
- These reactions usually last 6–12 days and then subside after the offending agent is eliminated.

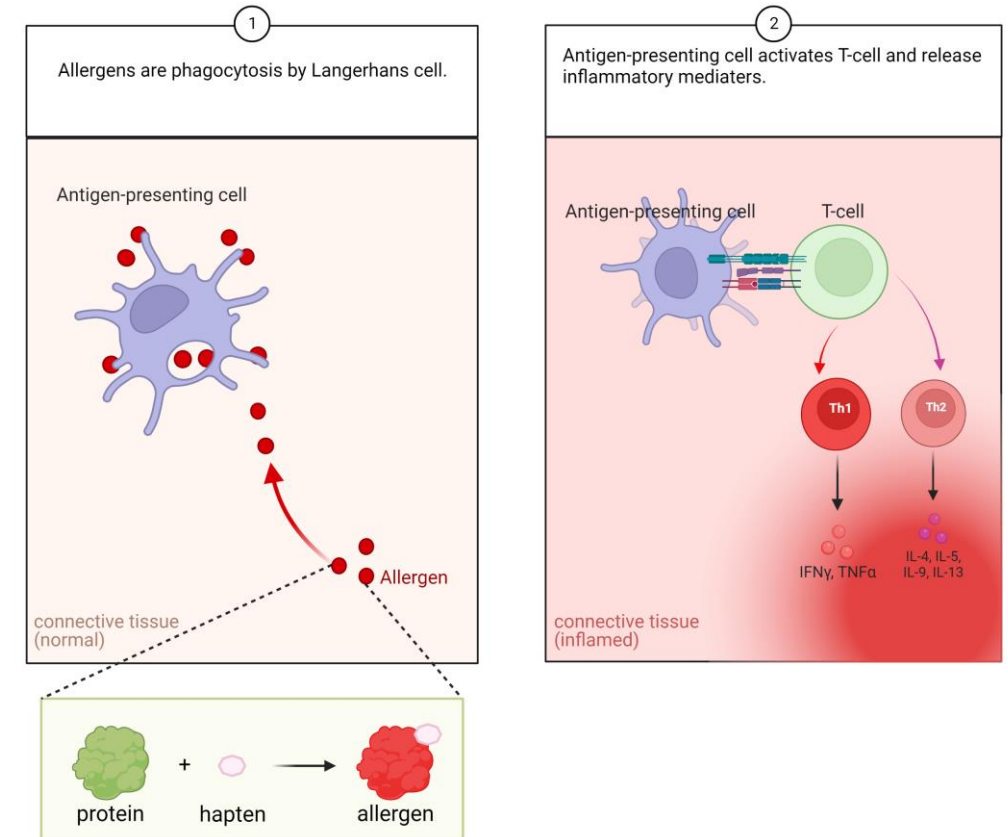




Type IV: Delayed hypersensitivity reaction

Delayed hypersensitivity reactions is mediated by sensitized T lymphocytes and macrophages.

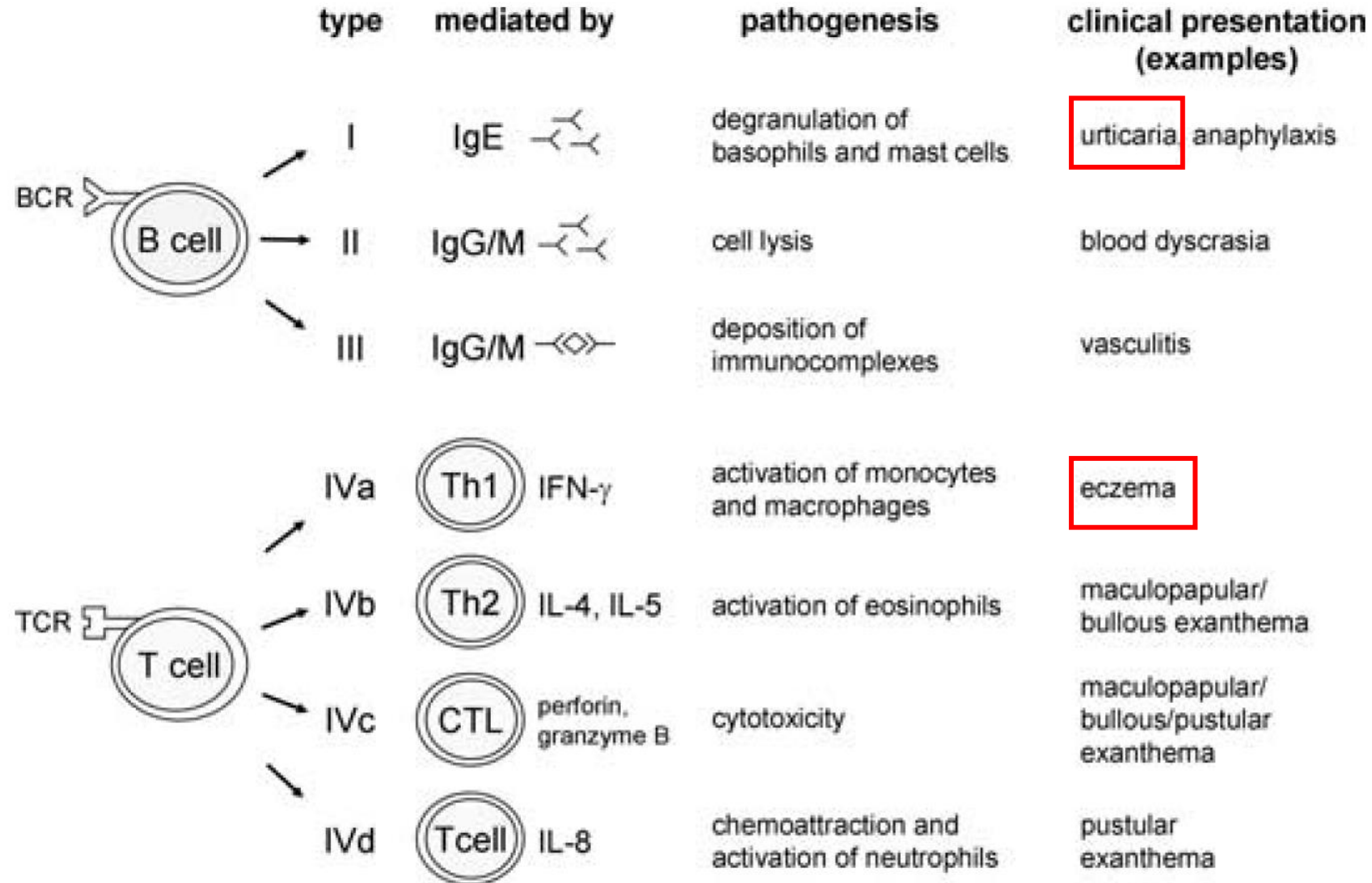
- The sensitized cells come in contact with antigen, an inflammatory reaction is generated by the production of
 - lymphokines
 - subsequent influx of neutrophils and macrophages.
- Several drugs, including commonly used antibiotics, can induce delayed hypersensitivity reactions, **sulfamethoxazole**, **ciprofloxacin**,





Skin rash VS hypersensitivity subtype

Classification of reactions responsible for immune-mediated hypersensitivity to drugs





Urticaria vs Eczema



Urticaria/Hives

<https://en.wikipedia.org/wiki/Hives>



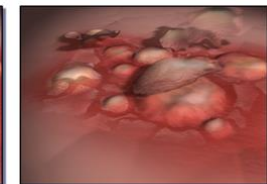
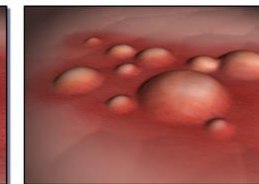
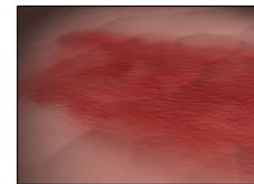
Eczema



Redness

Blisters

Flaking



Eczema

<https://en.wikipedia.org/wiki/Dermatitis>



Stevens-Johnson syndrome

The severe skin allergic.

- **Stevens-Johnson syndrome (SJS) (less than 10% of skin that effected)**
- **Toxic epidermal necrolysis (TEN) (more than 30% of skin that effected)**

Early symptoms of SJS

- Fever and flu-like symptoms
- A few days later, blister and peel of skin
- Mucous membrane symptom occurred
- Complication
 - Dehydration, sepsis, pneumonia, organ failure



SJS

https://en.wikipedia.org/wiki/Stevens%E2%80%93Johnson_syndrome

**Lamotrigine, carbamazepine, allopurinol,
sulfonamide, nevirapine**



Idiosyncratic Reactions and Pharmacogenetic Contributions

Idiosyncrasy is an abnormal reactivity to a chemical that is peculiar to a given individual

- the idiosyncratic response may be extreme sensitivity to low doses or extreme insensitivity to high doses of drugs.

A common mechanism is covalent drug binding to serum proteins that leads to the presentation of a foreign hapten, resulting in an immunotoxicological response. Many interindividual differences in drug responses have a pharmacogenetic basis.

- black males (~10%) develop a serious hemolytic anemia when they receive primaquine as an antimalarial therapy; this development is due to a genetic deficiency of erythrocyte glucose-6-phosphate dehydrogenase.
- Polymorphisms in NAT2 lead to a multimodal distribution of isoniazid acetylation and clearance.
- Variability in the anticoagulant response to warfarin is due to polymorphisms in CYP2C9 and VKORC1
 - In addition, CYP3A4 and CYP2D6 metabolize a large number of drugs in the liver.



Substances Most Frequently Involved in Human Poisoning Exposures

SUBSTANCE	%
Analgesics	11.3
Personal care products	7.7
Cleaning substances	7.7
Sedatives/hypnotics/antipsychotics	5.9
Antidepressants	4.4
Antihistamines	4.0
Cardiovascular drugs	4.0
Foreign bodies/toys/miscellaneous	3.9
Pesticides	3.2



Poisons Associated with the Largest Number of Human Fatalities

Sedatives/hypnotics/antipsychotics	Stimulants and street drugs
Cardiovascular drugs	Alcohols
Acetaminophen (alone and in combinations)	SSRIs
Opioids	



PRINCIPLES OF TREATMENT OF POISONING

- Maintain **vital** physiological **functions**
- Reduce or prevent **absorption** and enhance **elimination**
- **Combat the toxicological effects** of the poison at the effector sites

The “ABCDE” mnemonic of emergency care

Airway	Maintain patency
Breathing	Maintain oxygenation and ventilation
Circulation	Maintain perfusion of vital organs
Disability	Assess for CNS dysfunction, If neurological disability is noted, check -O ₂ administration (check pulse oximetry) -Dextrose administration (check blood sugar) -Naloxone administration (empirical trial) -Thiamine (for adult patients receiving dextrose)
Exposure	Assess “toxidrome”



PRINCIPLES OF TREATMENT OF POISONING: **Toxidrome**

Common Toxidromes

Groups of physical signs and symptoms associated with specific poisoning syndromes are known as **toxidromes**.

DRUG CLASS	EXAMPLE(S)	MENTAL STATUS	HR	BP	RR	T	PUPIL SIZE	OTHER
Sympathomimetic	Cocaine Amphetamine	Agitation	↑	↑		↑	↑	Tremor, diaphoresis
Anticholinergic	Diphenhydramine Atropine	Delirium	↑	↑		↑	↑	Ileus, flushing
Cholinergic	Organophosphates	Somnolence Coma			↑		↓	SLUDGE, ^a fasciculation
Opioid	Heroin Oxycodone	Somnolence Coma	↓		↓		↓	
Sedative-hypnotic	Benzodiazepines Barbiturates	Somnolence Coma		↓	↓			
Salicylate	Aspirin	Confusion	↑		↑	↑		Diaphoresis, vomiting
Ca ²⁺ channel blocker	Verapamil		↓	↓				

BP, blood pressure; HR, heart rate; RR, respiratory rate; T, temperature.

^a

SLUDGE, muscarinic effects of salivation, lacrimation, urination, defecation, gastric cramping, and emesis.



Toxicity testing in animal experiment

- *Acute exposure*—**exposure generally <24 h**
 - considered to be a single exposure (or dose)
 - may consist of repeated exposures within a short time period.
- *Subacute exposure*—duration is greater
 - from **several days to 1 month** in animal studies
- *Sub-chronic exposure*—exposures repeated or spread over an intermediate time range
 - for animal testing, **generally to be 1–3 months**
- *Chronic exposure*—exposures (repeated or continuous) over a long period of time
 - In animal testing: exposure ranges **between 90 days to a lifetime.**

- *Acute toxicity*—
 - ADR that is manifested within a relatively short time
 - almost immediately to within several days following exposure
 - asphyxiation from exposure to a high concentration of carbon monoxide (CO)
- *Chronic toxicity*—
 - a permanent or lasting adverse effect that is manifested after exposure to a toxicant
 - development of silicosis following a long-term exposure to silica



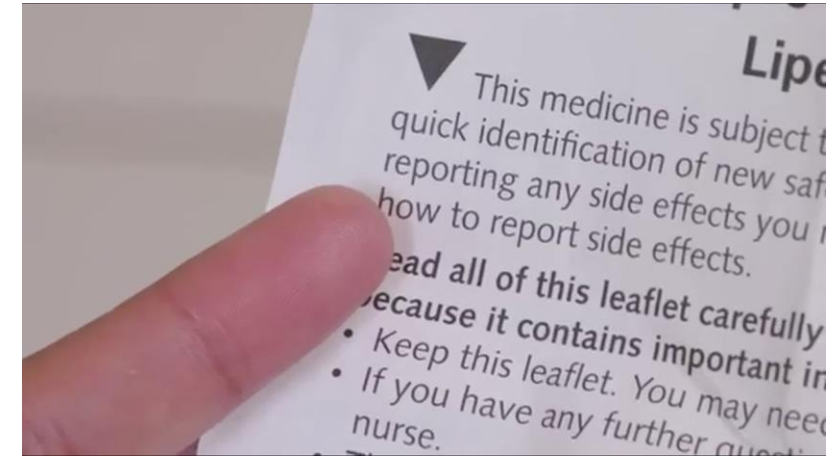
New Drug development

- การค้นพบ/สังเคราะห์/สกัด สารเคมีที่มีคุณสมบัติจะเป็นยาใหม่
- **In vitro** การศึกษาคุณสมบัติของยาใหม่ในหลอดทดลอง หรือในเซลล์เพาะเลี้ยง
- **In vivo** การศึกษาคุณสมบัติทางเภสัชวิทยา เภสัชจลนศาสตร์ เภสัชพลศาสตร์ และศึกษาความเป็นพิษของยาใหม่ในสัตว์ทดลอง
- เสนอโครงการเพื่อการทดสอบในคน
 - **Phase I** หาข้อมูลด้านความปลอดภัย และ เภสัชจลนศาสตร์ในอาสาสมัครปกติจำนวนน้อย (20-100 คน)
 - **Phase II** หาข้อมูลด้าน **efficacy, safety, dose** ในผู้ป่วยจำนวนปานกลาง (100-300 คน)
 - **Phase III** เก็บข้อมูล **efficacy, safety** ในประชากรจำนวนมาก (1,000-5,000 หรือมากกว่า)



Black triangle drugs ▼

- ▼ indicates that the CHM/MHRA are intensively monitoring that product
 - “This medicinal product is subject to additional monitoring.”
- ▼ will be assigned to a product because:-
 - the drug is new
 - the drug is being administered to the patient either by a **new route of administration** or a **new formulation** which is considered **may have an impact on** the already established risk/benefit profile of that drug
 - the drug is being administered for a **new indication**



<https://www.ema.europa.eu/en/human-regulatory/post-authorisation/pharmacovigilance/medicines-under-additional-monitoring>

การติดตามความปลอดภัยของยาใหม่ (Safety Monitoring Program : SMP) ตามหลักเกณฑ์การขึ้นทะเบียนตำรับยาใหม่ของสำนักงานคณะกรรมการอาหารและยาได้กำหนดให้มี การอนุมัติทะเบียนยาใหม่ โดยให้จำกัดการจำหน่ายใช้เฉพาะสถานพยาบาล (ทั้งภาครัฐและเอกชน) ที่มีแพทย์ ดูแลอย่างใกล้ชิดและมีการติดตามความปลอดภัยเป็นเวลา 2 ปีและให้บริษัทดำเนินการติดตามและเฝ้าระวัง ความปลอดภัย ซึ่งรายงานโดยแพทย์ทั่วไป, แพทย์เฉพาะทาง, เภสัชกรและพยาบาล โดยการรวบรวมและ จัดส่งรายงานอาการไม่พึงประสงค์จากการใช้ยาใหม่ให้สำนักงานคณะกรรมการอาหารและยา

ในระยะ 2 ปีแรกให้รายงานทุก 4 เดือน หลังจากที่ได้รับใบสำคัญการ ขึ้นทะเบียนตำรับยาแบบมีเงื่อนไข ยาใหม่ที่อยู่ระหว่างการติดตามความปลอดภัย ซึ่งยากลุ่มนี้จะมีสัญลักษณ์เป็นรูปสามเหลี่ยมและมี อักษรคำว่า “ต้องติดตาม” อยู่ภายในสามเหลี่ยม และได้สามเหลี่ยมจะเขียนคำว่า “ใช้เฉพาะสถานพยาบาล ยาควบคุมพิเศษ



http://sara-pan-ya.blogspot.com/2011/05/blog-post_8319.html



History of Pregnancy labeling

1979—FDA issues first labeling requirement—pregnancy labeled under “Precautions” section

1994—FDA forms Pregnancy Labeling Task Force after Teratology Society publishes position paper on pregnancy letter categories

1997—Public hearing convened to discuss letter categories

1999—FDA publishes white paper outlining changes, seeks more assistance from clinicians and others

2008—Proposed PLLR is published

2014—Final PLLR is published

Abbreviations: FDA=US Food and Drug Administration,
PLLR=Pregnancy and Lactation Labeling Rule.



Pregnancy labeling: new VS. old

Section	Labeling Prior to the PLLR	New Labeling System
8.1	Pregnancy	Pregnancy (includes Labor and Delivery)
8.2	Labor and Delivery	Lactation
8.3	Nursing Mothers	Females and Males of Reproductive Potential (new section)
	Categories A, B, C, D, X	Categories removed
Abbreviation: PLLR = Pregnancy and Lactation Labeling Rule.		



Pregnancy and lactation labeling (original)

The original FDA regulations on pregnancy, labor and delivery, and lactation labeling for prescription drugs were formulated in 1979. Currently, this risk assessment is presented in the commonly known form of pregnancy categories consisting of A, B, C, D, and X designations as described in this table.

Category	Description
A	Well-controlled studies in humans show no risk to the fetus
B	No well-controlled studies have been conducted in humans; animal studies show no risk to the fetus
C	No well-controlled studies have been conducted in humans; animal studies have demonstrated an adverse effect on the fetus
D	Evidence of human risk to the fetus exists; however, benefits may outweigh risks in certain situations
X	Controlled studies in animals or humans demonstrate fetal abnormalities; the risk in pregnant women clearly outweighs any possible benefit

Many limitations to this system were quickly identified and the FDA began public hearings in 1997 to evaluate the labeling regulations. Some of these limitations include:

- Purpose was only to present adequate information to assist when prescribing for patients already pregnant, not following drug exposure during an unplanned pregnancy
- The meaning of these categories is commonly misunderstood and inappropriately applied in practice
- Oversimplification leads to a misinterpretation
- 65% to 70% of all prescription drugs are assigned to Pregnancy Category C yet have drastically different sets of supporting evidence and risks
- Lack of differentiation between data from animal and human studies



Pregnancy and lactation labeling (new-2015)

Subsection	Description
Pregnancy Exposure Registry	Scientifically acceptable registry and contact information
Risk Summary	Known risks in context with background rates of adverse events in general and (if possible) in disease populations
Clinical Considerations	Medical/disease factors that should be considered
Data	Data that support the risk summary, with human data highlighted first
Abbreviation: PLLR = Pregnancy and Lactation Labeling Rule.	

Rate of major malformations in general population: 3%–4%²
Rate of premature delivery: 11%–12%³
Rate of gestational diabetes: 2%–7%⁴
Untreated psychiatric disorders carry risks for woman and baby⁵
Obesity increases obstetric risks⁶

Table 2. Pregnancy Subsection: Overview

General Information

Contact information if pregnancy registry available
General statement about background risk

Fetal Risk Summary

Based on all available data, this section characterizes the likelihood that the drug increases the risk of developmental abnormalities in humans and other relevant risks. More than one risk conclusion may be needed
For drugs that are not systemically absorbed, there is a standard statement indicating that maternal use is not expected to result in fetal exposure

For drugs that are systemically absorbed, it will include:

- Where there are human data, a statement about the likelihood of increased risk based on these data. This statement is followed by a description of findings
- A standard statement about likelihood of increased risk based on animal data

Clinical Considerations

This section provides information on the following topics:

- Inadvertent exposure
 - Known or predicted risk to the fetus from inadvertent exposure to drug early in pregnancy
- Prescribing decisions for pregnant women
 - Description of any known risk to the pregnant woman and fetus from the disease or condition the drug is intended to treat
 - Information about dosing adjustments during pregnancy
 - Maternal adverse reactions unique to pregnancy or increased in pregnancy
 - Effects of dosage, timing, and duration of exposure to drug during pregnancy
 - Potential neonatal complications and needed interventions

Data

Human and animal data are presented separately with human data presented first:

- Description of study type, exposure information (dosage, duration, timing), and any identified fetal developmental abnormality or other adverse effects
- For human data, it will include positive and negative experiences, number of subjects, and duration of study
- For animal data, it will include species studied and description of dosages in terms of human dosage equivalents



Pregnancy and lactation labeling (new-2015)

Table 3. Lactation Subsection: Overview

Risk Summary

For drugs that are not systemically absorbed, there is a standard statement indicating that maternal use is not expected to result in infant exposure

For drugs that are systemically absorbed, the risk summary must be a description of the following information or it will state that it is not available:

- Effects of drug on milk production
- Presence of drug in human milk
 - If drug not detected, it will state limits of assay
 - If drug is detected, provide drug concentration in milk and estimated infant daily dose (actual and compared to pediatric or maternal doses)
 - Effects of the drug on the breastfed child
- If data show that the drug does not affect the quantity and quality of breast milk and there is reasonable certainty that either the drug is not detectable in breast milk or will not adversely affect the breastfed child, then it will state:
 - “The use of [name of drug] is compatible with breastfeeding.”

Clinical Considerations

This section must provide, when available, information on:

- Ways to minimize exposure of the breastfed infant to the drug
- Dosing adjustments during lactation

Data

This section must provide an overview of the data that are the basis for information in the risk summary and clinical considerations sections